CauESC with Persona-Attention and Two-Stage Reinforcement Learning for Emotional Support Conversation

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Abstract

Emotional support conversation systems require empathetic understanding, commonsense reasoning, and strategic response generation. We present an enhanced CauESC architecture that integrates (1) COMET-derived commonsense knowledge across five causal dimensions, (2) a Persona Attention Loop (PAL) for faithful persona conditioning, (3) a multi-factor decoder with dual cross-attention over encoder states and strategy executors, and (4) two-stage training combining pointer-generator networks with reinforcement learning optimized for ROUGE-L and BLEU-2 rewards. On the ESConv test set, our system achieves BLEU-4 of 3.92, ROUGE-L of 16.91, METEOR of 17.37, and perplexity of 11.41, demonstrating improved fluency and overlap metrics compared to the Stage 1 baseline. We conduct a comprehensive ablation study removing PAL, the multi-factor decoder, and RL training, confirming that each component contributes to performance gains. Our code, trained models, and reproducibility instructions are publicly available.

CCS Concepts

• **Computing methodologies** → **Natural language generation**; **Discourse, dialogue and pragmatics**; *Neural networks.*

Keywords

emotional support conversation, reinforcement learning, COMET, persona conditioning, strategy prediction, empathetic dialogue

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1 Introduction

Mental health challenges affect millions worldwide, yet access to professional counseling remains limited. Emotional support conversation (ESC) systems aim to provide empathetic dialogue that helps users cope with stress, anxiety, and other emotional difficulties. Unlike task-oriented dialogue systems, ESC requires understanding user affect, leveraging background knowledge about situations and emotions, and selecting appropriate counseling strategies such as questioning, reflection, or providing suggestions.

The ESConv dataset [1] provides annotated dialogues between help-seekers and supporters, labeled with eight counseling strategies. Building on this foundation, CauESC [2] introduced causal reasoning by incorporating COMET [3] commonsense knowledge and emotion-cause annotations from RECCON [4]. However, the original CauESC implementation lacks explicit persona conditioning mechanisms and does not employ reinforcement learning to optimize generation quality.

1.1 Project Goals and Phase 3 Requirements

This project extends CauESC with three major architectural innovations and a comprehensive evaluation framework to satisfy Phase 3 course requirements:

(1) **Persona Attention Loop (PAL):** A co-attention mechanism that jointly encodes persona descriptions and dialogue history, ensuring generated responses align with supporter personas.
(2) **Multi-factor decoder:** Enhanced BlenderBot-small decoder layers with dual cross-attention—one over encoder states and another over strategy executor embeddings—fused via learned gates.
(3) **Two-stage training:** Stage 1 pretrains a pointer-generator network (PGN) with maximum likelihood estimation; Stage 2 applies mixed reinforcement learning with rewards based on ROUGE-L and BLEU-2 scores.
(4) **Reproducible infrastructure:** Deterministic beam search inference, automated metric logging including NLGEval suite, confusion matrix tracking, and summary report generation.

1.2 Individual Contributions

Author 1 implemented the Persona Attention Loop architecture, integrated the two-stage training pipeline with RL reward computation, developed the deterministic inference framework, and created the automated summary generation script. **Author 2** managed COMET knowledge extraction and preprocessing, configured ablation experiments across different model variants, implemented the comprehensive evaluation harness including NLGEval integration, and drafted documentation including the experiment plan and README files.

*Both authors contributed equally to this research.

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1.3 Paper Organization

The remainder of this paper is organized as follows: Section 2 reviews related work in emotional support dialogue, commonsense reasoning, and strategy-aware generation. Section 3 provides background on the ESConv dataset and COMET knowledge framework. Section 4 details our architectural innovations including PAL, the multi-factor decoder, and training procedures. Section 5 describes experimental setup and resources. Section 6 presents quantitative metrics, qualitative analysis, and our mandatory ablation study. Section 7 discusses limitations and future work, and Section 8 concludes.

2 Related Work

2.1 Emotional Support Conversation

The ESConv dataset [1] introduced 1,053 dialogues with strategy annotations across eight categories: Question, Restatement or Paraphrasing, Reflection of feelings, Self-disclosure, Affirmation and Reassurance, Providing Suggestions, Information, and Others. Early baselines including Transformer, Multi-Task Transformer, MoEL [5], and MIME [6] achieved limited success in strategy prediction and empathetic response generation.

BlenderBot-Joint and DialoGPT-Joint finetune pretrained models on ESConv but lack explicit strategy modeling. MISC [7] introduced multi-grained positive sampling but does not incorporate external commonsense knowledge. CauESC [2] addresses this gap by integrating COMET features and causal attention mechanisms, achieving state-of-the-art results with 18.20 ROUGE-L and 8.17 BLEU-2 on the test set.

2.2 Commonsense and Emotion Cause Reasoning

COMET [3] generates commonsense inferences about events, producing if-then knowledge such as "If X wants to comfort Y, then X needs to understand Y's feelings." COSMIC [8] applies COMET to dialogue understanding, extracting mental state representations for emotion recognition.

RECCON [4] focuses on recognizing emotion causes within conversations, annotating which utterances trigger specific emotional responses. CauESC combines these two frameworks, using COMET to generate situation, seeker, and supporter mental states while leveraging RECCON annotations to apply causal attention masks.

2.3 Strategy Modeling and Reinforcement Learning

Strategy prediction in counseling dialogue has been approached through auxiliary classification tasks [9] and strategy-conditioned generation [10]. However, most prior work treats strategy as a discrete label rather than integrating it deeply into the generation process.

Reinforcement learning has been applied to general dialogue systems to optimize engagement [11] and information gain [12], but applications to emotional support remain limited. Our work combines supervised pretraining with RL fine-tuning, using lexical overlap rewards (ROUGE-L + BLEU-2) while tracking strategy accuracy as a diagnostic metric.

3 Background

3.1 ESConv Dataset

ESConv provides 1,053 multi-turn dialogues between help-seekers expressing emotional distress and trained supporters offering guidance. Each dialogue includes:

- **Persona:** Short descriptions of seeker background (e.g., "my job was lost due to pandemic")
- **Context:** Multi-turn dialogue history
- **Response:** Supporter utterance
- **Strategy:** One of eight counseling strategies
- **Emotion:** Seeker emotional state (32 fine-grained labels)

We use the official train/valid/test splits containing 720/91/242 dialogues respectively. Text is preprocessed via `_reformat/process.py`, producing cleaned files like `test_clean_final.txt` with empty responses removed.

3.2 COMET Knowledge Streams

Following CauESC, we extract five types of commonsense knowledge:

(1) K^{ES}: Situation mental states (e.g., "wants to find a job")
(2) K^{EC}_{intra}: Seeker utterance effects (e.g., "feels stressed")
(3) K^{EC}_{inter}: Supporter utterance effects (e.g., "wants to help")
(4) Executor: Agent performing actions
(5) Persona: Background attributes

These features are precomputed using the COSMIC extraction pipeline and stored in `data/comet.<config>/<timestamp>/` directories. Configuration files like `comet_without_cause.json` allow selective ablation of knowledge streams.

4 Methodology

4.1 Architecture Overview

Our system architecture extends CauESC with PAL and enhanced decoder components, integrating COMET knowledge, cause-aware encoding, multi-factor decoding with strategy executors, and two-stage training with MLE and RL objectives.

4.2 Data Preprocessing Pipeline

Step 1: Text Processing. The `_reformat/process.py` script tokenizes persona sentences, dialogue context (up to 150 tokens), and responses (up to 50 tokens) using the BlenderBot-small tokenizer. It aligns strategy labels and emotion annotations, producing train/valid/test JSON files.

Step 2: COMET Extraction. The `scripts/prepare_comet.sh` bash script invokes COSMIC feature extraction tools, generating commonsense inferences for each dialogue turn. Outputs are cached as PyTorch tensors in the data directory.

Step 3: Batch Construction. The `inputters/comet.py` module loads preprocessed data and assembles batches containing:

- Input IDs and attention masks (max length 150)
- Decoder input IDs (max length 50)
- COMET knowledge embeddings (5 streams × sequence length)
- Strategy labels (8 classes)
- Persona embeddings

Configuration flags like `-use_all_persona` and `-encode_context` control which features are included, enabling ablation experiments.

4.3 Cause-Aware Encoder

The encoder builds on BlenderBot-small (90M parameters) with the following enhancements:

Self-Attention. Standard bidirectional self-attention over concatenated persona and dialogue tokens, producing encoder states H^Q (situation-aware), H^C (context-aware), and H^S (strategy history).

Cause Attention. Emotion cause masks derived from RECCON annotations restrict attention to utterances that triggered the seeker's current emotional state. This causal filtering improves the model's understanding of why users feel distressed.

4.4 Persona Attention Loop (PAL)

PAL implements bidirectional co-attention between persona representations H_P and dialogue history H_D:

$$\begin{aligned} H'_P &= \text{CoAttn}(H_P, H_D) + H_P \\ H'_D &= \text{CoAttn}(H_D, H_P) + H_D \end{aligned} \tag{1}$$

where CoAttn denotes scaled dot-product cross-attention with residual connections. The updated representations are combined to produce decoder inputs.

Final logits are computed as:

$$\text{logits} = (1 + \alpha) \cdot \text{logits}_{P+D} - \alpha \cdot \text{logits}_D \tag{2}$$

with $\alpha = 0.3$ emphasizing persona-conditioned outputs. This formulation encourages the model to reference persona details (e.g., mentioning the user's job situation or family support) when generating responses.

4.5 Multi-Factor Decoder with Strategy Executors

The decoder, implemented in `models/multi_factor_decoders.py`, modifies each BlenderBot-small decoder layer to include:

Dual Cross-Attention. Two separate cross-attention modules:

(1) Standard encoder cross-attention over H^C and H^Q
(2) Strategy executor cross-attention over strategy embeddings H^{strat}

Fusion Gate. A learned gating mechanism combines the two attention outputs:

$$\begin{aligned} g &= \sigma(W[h; h_{\text{enc}}; h_{\text{strat}}]) \\ h_{\text{fused}} &= g \odot h_{\text{enc}} + (1 - g) \odot h_{\text{strat}} \end{aligned} \tag{3}$$

where h is the decoder hidden state, W is a learnable weight matrix, σ is the sigmoid function, and $\odot$ denotes element-wise multiplication. This allows the model to dynamically balance causal context and strategy guidance at each decoding step.

Pointer-Generator Network. Following See et al. [13], we maintain a copy distribution over input tokens alongside the vocabulary distribution, enabling the model to reproduce factual details from the dialogue context or persona descriptions.

4.6 Strategy Predictor

A classification head over H^S predicts strategy logits $\hat{s} \in \mathbb{R}^8$. During training, we compute cross-entropy loss against ground-truth labels. During inference, predicted strategies condition the decoder via strategy executor embeddings. We track strategy accuracy and confusion matrices to diagnose model behavior.

4.7 Two-Stage Training

Stage 1: Pointer-Generator Pretraining. We train the full architecture with maximum likelihood estimation:

$$\mathcal{L}_{\text{MLE}} = -\sum_{t=1}^{T} \log P(y_t | y_{<t}, x) \tag{4}$$

plus auxiliary strategy classification loss with weight 0.5. Training uses AdamW optimizer with learning rate 5×10^{-6}, batch size 8, and gradient clipping at norm 1.0. We train for 2 epochs with validation every 1000 steps. The best checkpoint based on validation perplexity is saved as `checkpoints/stable_pgn_model.pth`.

Stage 2: Mixed Reinforcement Learning. We resume from the Stage 1 checkpoint and introduce a mixed objective:

$$\mathcal{L}_{\text{mixed}} = (1 - \alpha)\mathcal{L}_{\text{MLE}} + \alpha \mathcal{L}_{\text{RL}} \tag{5}$$

where $\alpha = 0.1$ controls the mixing ratio. The RL loss uses REINFORCE [14]:

$$\mathcal{L}_{\text{RL}} = -\mathbb{E}_{y \sim \pi_\theta}[R(y, y^*) \log \pi_\theta(y|x)] \tag{6}$$

with reward function:

$$R(y, y^*) = 0.6 \cdot \text{ROUGE-L}(y, y^*) + 0.4 \cdot \text{BLEU-2}(y, y^*) \tag{7}$$

We compute ROUGE-L using the Python `rouge` library and BLEU-2 via NLTK. Rewards are computed per mini-batch and normalized by subtracting a baseline (moving average of past rewards) to reduce variance.

Early stopping monitors weighted validation metric ($0.6 \cdot$ROUGE-L$+ 0.4 \cdot$BLEU-4), selecting the checkpoint with the highest score. This best checkpoint is saved as `best_generation.bin`.

4.8 Inference Configuration

To ensure reproducibility, we enforce deterministic beam search decoding:

- Number of beams: 4
- Length penalty: 1.5
- Repetition penalty: 1.2
- No repeat n-gram size: 0
- Temperature: 1.0 (not used with beams)
- Top-k: 0 (disabled)
- Top-p: 1.0 (disabled)
- Min length: 5 tokens
- Max length: 50 tokens

All experiments use `-use_all_persona True` and `-encode_context True` to activate persona and dialogue history encoding. The `-add_nlg_eval` flag triggers NLGEval metric computation.

5 Experimental Setup

5.1 Hardware Resources

All experiments were conducted on a shared research cluster with the following specifications:

- **GPU:** NVIDIA RTX 6000 Ada Generation with 24 GB GDDR6 memory
- **CPU:** Dual-socket Intel Xeon Silver 4516Y+ processors (2 × 24 cores, 48 threads total) with base clock 2.2 GHz and turbo boost up to 3.7 GHz
- **RAM:** 125 GiB DDR4 system memory (115 GiB available during experiments)
- **Swap:** 119 GiB swap space
- **Storage:** NVMe SSD for dataset caching and checkpoint storage

GPU utilization was queried via `nvidia-smi`, CPU specifications via `lscpu`, and memory status via `free -h`.

5.2 Training Time and Computational Cost

Stage 1 Training: Ran for 2 epochs over the 720-dialogue training set with batch size 8 and gradient accumulation 1, totaling approximately 180 update steps per epoch. Wall-clock time: 6 hours on the RTX 6000 Ada.

Stage 2 Training: Continued from the Stage 1 checkpoint for 2 additional epochs with mixed MLE-RL loss. RL reward computation added overhead due to batch-level ROUGE/BLEU scoring. Wall-clock time: 8 hours.

Total Training Time: 14 GPU-hours. Assuming 300W GPU power draw, estimated energy consumption is approximately 4.2 kWh.

5.3 Evaluation Protocol

We evaluate on the official ESConv test set (`_reformat/test_clean_final.txt`) containing 242 dialogues with 1,728 total turns. Inference uses batch size 16 with deterministic beam search as described in Section 4.7.

Automatic Metrics:

- BLEU-1/2/3/4 [15]
- ROUGE-L [16]
- METEOR [17]
- Distinct-1/2/3 [18] measuring lexical diversity
- Perplexity
- Strategy accuracy (top-1 and top-3)
- Macro/micro F1 for strategy classification

NLGEval Suite: We additionally compute CIDEr, embedding-based metrics (cosine similarity, greedy matching, vector extrema) using pre-loaded GloVe 6B 300d embeddings via gensim 4.3.3.

Confusion Matrix: We track an 8 × 8 confusion matrix for strategy predictions to diagnose systematic errors.

5.4 Automated Reporting

After each inference run, we execute a Python script that generates `*_summary.txt` files containing:

(1) Run timestamp and checkpoint path
(2) Complete inference arguments (JSON format)
(3) All automatic metrics
(4) Strategy confusion matrix
(5) First 15 generated outputs with ground truth and strategy labels

This infrastructure enables rapid comparison across experiments and ensures all results are archived.

6 Results

6.1 Main Results

Table 1 compares our Stage 1 (PGN baseline) and Stage 2 (PGN + RL) models on the ESConv test set.

Table 1: Automatic metrics on ESConv test set. Best results in bold.

Metric	Stage 1	Stage 2 (Ours)
BLEU-1	13.15	**18.36**
BLEU-2	6.34	**9.00**
BLEU-3	3.93	**5.69**
BLEU-4	2.66	**3.92**
ROUGE-L	15.76	**16.91**
METEOR	13.22	**17.37**
Dist-1	**3.03**	1.66
Dist-2	**21.77**	13.73
Dist-3	**38.97**	25.30
Perplexity	11.95	**11.41**
Length (tokens)	17.63	34.91
Strategy Acc (top-1)	0.334	**0.336**
Strategy Acc (top-3)	0.615	**0.599**
Strategy F1 (macro)	0.217	0.208
Strategy F1 (micro)	0.334	0.336

Key Observations:

(1) Stage 2 achieves substantial gains in BLEU scores (+5.21 BLEU-1, +2.66 BLEU-2, +1.26 BLEU-4) and ROUGE-L (+1.15), validating the RL objective.
(2) METEOR improves significantly (+4.15), indicating better synonym matching and stemming.
(3) Perplexity decreases slightly (−0.54), suggesting improved fluency.
(4) Diversity metrics (Dist-1/2/3) drop substantially, a known side effect of beam search and RL optimization toward specific n-gram overlap targets.
(5) Average response length nearly doubles (17.63 → 34.91 tokens), likely due to length penalty 1.5 encouraging longer outputs.
(6) Strategy accuracy remains nearly constant, with macro F1 slightly decreasing, indicating the RL reward does not directly optimize strategy correctness.

6.2 Strategy Confusion Analysis

The confusion matrix for Stage 2 predictions reveals systematic bias toward class 5 (Affirmation and Reassurance), which is the most frequent strategy in the training set. Classes 1, 2, and 6 receive zero predictions, indicating complete failure to generalize to these

strategies. This suggests the RL reward, which only measures lexical overlap, does not provide sufficient signal to encourage strategy diversity.

6.3 Qualitative Analysis

We manually inspected the first 15 generated responses in the summary file. Representative examples:

Success Case:

> *User:* "I have tried meditation before, but quite a bit back. Do you have any activities you could recommend, without risking my health?"
> *Ground Truth:* "one exercise i find helpful is lying on my back with a pillow under my head..."
> *Generation:* "that sounds like a great idea! i'm glad to hear that you're feeling a bit better now. is there anything else you'd like to talk about today?"
> *Analysis:* The model produces a fluent, empathetic response and attempts to continue the conversation. However, it fails to provide the specific suggestion (breathing exercise) in the ground truth. True strategy: Providing Suggestions (class 5); Predicted: Affirmation (class 5).

Failure Case:

> *User:* "Thank you, I think I am. I am always worrying about my mother, so this is nothing new. Although, financially this will be more stressful."
> *Ground Truth:* "if i am understanding you correctly, you feel your mom may not be able to sustain herself during her time of recovery, is that correct?"
> *Generation:* "it sounds like you are worrying about your mom, so this is nothing new. i am always worrying about my mom, so this is nothing new. although, financially this will be more..."
> *Analysis:* The model copies verbatim from the user input and trails off incompletely. True strategy: Restatement (class 1); Predicted: Question (class 0). This demonstrates a pointer-generator failure mode where excessive copying produces incoherent outputs.

Overall, generations are grammatically correct and display persona awareness (referencing family, jobs, etc.) but often resort to generic reassurance rather than executing specific strategies like asking clarifying questions or offering concrete suggestions.

6.4 Ablation Study

To validate our architectural choices, we trained three ablation variants:

(1) **No PAL:** Set `-use_all_persona False` during training and inference, removing persona conditioning.
(2) **Vanilla Decoder:** Replace multi-factor decoder with standard BlenderBot-small decoder (config: `comet.json` instead of `comet_with_multi_factor_decoder.json`).
(3) **No RL (Stage 1 Only):** Use the Stage 1 checkpoint without RL fine-tuning.

Results are shown in Table 2.

Table 2: Ablation study results on ESConv test set.

Model Variant	BLEU-2	ROUGE-L	Dist-2	Strat Acc
Full Model (Stage 2)	**9.00**	**16.91**	13.73	**0.336**
-- No PAL	7.84	15.62	**14.85**	0.332
-- Vanilla Decoder	6.95	15.01	12.10	0.301
-- No RL (Stage 1)	6.34	15.76	**21.77**	0.334

Analysis:

- Removing PAL decreases BLEU-2 by 1.16 and ROUGE-L by 1.29, confirming that persona conditioning improves lexical overlap with references.
- Switching to vanilla decoder reduces BLEU-2 by 2.05 and strategy accuracy by 0.035, demonstrating the value of dual cross-attention and fusion gates.
- Removing RL (using only Stage 1) decreases BLEU-2 by 2.66 while *increasing* Dist-2 by 8.04, highlighting the diversity-fluency tradeoff introduced by RL optimization.

All three components contribute positively to overlap-based metrics, justifying our architectural design.

7 Discussion

7.1 Limitations

Strategy Collapse. The confusion matrix reveals severe class imbalance in predictions. Classes 1 (Restatement), 2 (Reflection), and 6 (Information) are never predicted, while class 5 (Affirmation) is overused. This stems from two factors: (1) the RL reward function optimizes ROUGE/BLEU without strategy supervision, and (2) class 5 is the most frequent in training data. Future work should incorporate strategy-specific rewards or use weighted cross-entropy loss during RL.

Diversity Reduction. Beam search with lexical overlap rewards favors safe, repetitive phrasing. Distinct-2 drops from 21.77 to 13.73. Exploring top-p sampling or diverse beam search could restore lexical variety while maintaining fluency.

Evaluation Metrics. BLEU and ROUGE measure n-gram overlap but do not capture empathy, helpfulness, or safety. Human evaluation studies are needed to assess whether generated responses are appropriate in sensitive counseling contexts.

Computational Cost. RL training requires 8 GPU-hours beyond the 6-hour MLE pretraining, limiting iteration speed. More efficient RL algorithms (e.g., proximal policy optimization) may reduce cost.

7.2 Ethical Considerations

Deploying emotional support systems raises significant ethical concerns:

- **Clinical Validity:** Our system is a research prototype and has not been validated by mental health professionals. It should not replace therapy or crisis counseling.
- **Safety:** The model may generate insensitive or harmful responses. Extensive safety testing and human-in-the-loop moderation are required before any real-world deployment.

- **Privacy:** ESConv dialogues are anonymized, but persona descriptions may contain identifying information. Future systems must implement robust privacy safeguards.
- **Bias:** The model may exhibit biases present in training data (e.g., demographic skew, cultural assumptions). Fairness audits are essential.

We emphasize that this work is purely academic and intended to advance NLP research, not to provide actual mental health services.

7.3 Future Directions

Several promising research directions emerge from this work:

(1) **Multi-Objective RL:** Extend the reward function to include strategy accuracy, empathy scores from trained classifiers, and safety constraints.
(2) **Hybrid Decoding:** Combine beam search for high-quality outputs with stochastic sampling to restore diversity.
(3) **Human Evaluation:** Conduct user studies with trained counselors to assess response quality, appropriateness, and empathy.
(4) **Multilingual Extension:** Adapt the approach to non-English emotional support datasets.
(5) **Real-Time Deployment:** Optimize inference speed and integrate with conversational interfaces for pilot studies under IRB approval.

8 Conclusion

We presented an enhanced CauESC architecture for emotional support conversation that integrates persona conditioning via PAL, multi-factor decoding with strategy executors, and two-stage training combining MLE and RL. On the ESConv test set, our system achieves BLEU-4 of 3.92, ROUGE-L of 16.91, and perplexity of 11.41, outperforming the Stage 1 baseline on overlap-based metrics. A comprehensive ablation study confirms that PAL, the multi-factor decoder, and RL training each contribute to performance gains.

However, significant challenges remain. Strategy predictions collapse to the majority class (Affirmation), and RL optimization reduces lexical diversity. Future work must address these issues through multi-objective rewards, hybrid decoding, and human evaluation.

This project satisfies Phase 3 requirements by delivering an 8-10 page ACM-formatted report, clearly stating individual contributions, documenting experimental resources (GPU, CPU, RAM, training time), conducting a mandatory ablation study, and ensuring reproducibility through comprehensive documentation.

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A Reproducibility Checklist

To ensure full reproducibility of our results, we provide:

(1) **Public Code Repository:** Complete source code including training scripts (`train.py`), inference scripts (`infer.py`), model definitions (`models/multi_factor_decoders.py`), data loaders (`inputters/comet.py`), and evaluation utilities (`eval_nlg_metrics.py`).

(2) **Configuration Files:** All experiment configurations stored as JSON files in `config/` directory, including `comet_with_multi_factor_decoder.json` for the full model and ablation variants.
(3) **Dependency Management:** `requirements.txt` lists all Python packages (PyTorch 2.1.0, transformers 4.34.0, datasets 2.15.0, gensim 4.3.3, nltk 3.8, rouge 1.0.1, nlgeval). `env.yml` provides conda environment specification.
(4) **Preprocessing Scripts:** `_reformat/process.py` for text cleaning and `scripts/prepare_comet.sh` for COMET feature extraction, with detailed inline documentation.
(5) **Training Commands:** Exact bash commands for Stage 1 and Stage 2 training documented in `docs/final_stage2_training.md`.
(6) **Inference Commands:** Complete inference command with all hyperparameters specified in Section 4.7.
(7) **Trained Checkpoints:** Best checkpoints saved at `checkpoints/stable_pgn_model.pth` (Stage 1) and `best_generation.bin` (Stage 2), with SHA-256 checksums.
(8) **Evaluation Outputs:** All metric files (`metric.json`, `metric_nlgeval.js` confusion matrices, generated texts (`gen.json`), and summary reports archived in `cauesc_result/` directory.
(9) **README Documentation:** Comprehensive `README.md` with project overview, environment setup, dataset preparation, training procedures, inference procedures, evaluation metrics, and ablation experiment commands.
(10) **Hardware Specifications:** Documented in Section 5 with exact CPU/GPU/RAM details.

B Sample Outputs

Table 3 shows three representative examples from the test set with ground truth responses and our model's generations.

These examples illustrate the model's tendency to ask questions (strategy 0) across diverse contexts, even when ground truth uses affirmation (4) or suggestions (3). However, responses remain empathetic and contextually appropriate.

Table 3: Sample outputs from ESConv test set. True/Pred columns indicate ground-truth and predicted strategy IDs.

User Post	Ground Truth	Generation	True	Pred
"Hi, I have been feeling a bit down lately. My levels of stress and concern have been higher since the pandemic began."	"sorry you have been feeling down. what are some strategies you have tried to help you cope?"	"i'm sorry to hear that. is there anything specific you'd like to discuss with me today?"	0	0
"Well, I have tried to get my mind off of it by activities at home. Spending time with my family."	"those sound like positive ways of handling stress. it is natural of course to feel stressed out by the pandemic so it can be useful to think of ways to help manage that stress."	"that is a great idea. activities can be very helpful. do you have any hobbies or activities that you enjoy doing with your family?"	4	0
"Yes, but I still seem to struggle with feeling a bit more down than I'd like. Not sure what to do."	"have you tried meditation and/or deep breathing exercises? i know i find that helpful when i am worrying."	"it sounds like you are struggling with feelings of sadness. is that correct?"	3	0